SCViT: A Spatial-Channel Feature Preserving Vision Transformer for Remote Sensing Image Scene Classification

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***Abstract*— Convolutional neural network (CNN)-based methods are widely used in remote sensing image scene classification and can obtain excellent performances. However, the stacked receptive fields in the CNN-based methods have limitations in modeling the long-range dependencies of local features. The vision transformer (ViT) model provides a good solution as it directly considers the global interactions of local patches by the self-attention mechanism. However, the vanilla ViT model, which simply splits images into fixed-size patches treated as tokens, mainly considers the global information in the spatial domain. In this article, a spatial-channel feature preserving ViT (SCViT) model is proposed, which considers both the detailed geometric information of the high-spatial-resolution (HSR) imagery and the contribution of the different channels contained in the classification token. First, in the proposed method, tokens are generated by progressively aggregating the neighboring overlapping patches to extract the local structural features of the imagery. Second, a multihead self-attention (MSA) mechanism is used to model the global interactions of the tokens in the encoder. A lightweight channel attention (LCA) module is then introduced to consider the importance of the different channels in the classification token. Finally, a multilayer perceptron (MLP) is used to acquire the final results. Compared with the state-of-the-art scene classification methods, the experimental results confirm the potential of using ViT models in remote sensing image scene classification.**

***Index Terms*— Attention mechanism, high spatial resolution (HSR), remote sensing image, scene classification, vision transformer (ViT).**

I. Introduction

In RECENT years, high-spatial-resolution (HSR) remote sensing images that contain abundant land-cover details have become more and more available. HSR images can be used in various applications, such as urban planning [1], natural hazard detection [2], and environmental monitoring [3]. The land-use/land-cover (LULC) types in the imagery can be recognized by the use of pixel- and object-based classification methods. However, these methods easily suffer from the spectral variability problem [4] and have difficulty in considering the ratio and spatial distribution of the different objects in HSR imagery, which is called the semantic gap problem [5], [6]. Scene classification is a technique that can be used to extract the high-level semantic information of HSR imagery and directly builds the relationship between the imagery and the functional area of the scene [7], [8].

The traditional scene classification methods are mostly based on handcrafted features. In the early years, visual attributes, such as color, texture, and edges, were used to construct low-level features [9]–[11]. However, low-level features, which are simple combinations of basic elements, have limitations in expressing the global information of a scene, and scene classification results obtained using low-level features are often unsatisfactory. To better extract the semantic information from the image content, mid-level features have been proposed, which are based on the encoding of the low-level features [8], [12]–[14]. However, the handcrafted features are usually designed based on prior knowledge, and the performance of the traditional scene classification methods needs to be further improved when dealing with a large amount of multisource remote sensing data containing complex distributions.

As a data-driven method, deep learning has been gradually used in remote sensing tasks in recent years [15] and achieves promising results, such as classification [16], [17], detection [18], unmixing [19], and segmentation [20]. In particular, convolutional neural network (CNN)-based methods have been widely used in scene classification. In 2015, Castelluccio *et al.* [21] introduced a fine-tuning strategy to CaffeNet and GoogLeNet for land-use classification with remote sensing data. Hu *et al.* [22] and Wang *et al.* [23] explored the use of CNN features as an input for mid-level classifiers for remote sensing image scene classification. In 2019, Zhang *et al.* [24] proposed an end-to-end scene classification method, in which the CNN features are fed into a capsule network (CapsNet) to obtain the final results.

To consider the relative importance of the different channels and spatial regions in the feature maps, attention mechanisms have been introduced into CNN architectures. For example, Zhu *et al.* [25] proposed a framework named the

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attention-based deep feature fusion (ADFF) method, which fuses the original images and Grad-CAM attention maps as the input to discriminate repeated textures and the salient regions; Wang *et al.* [26] proposed the end-to-end attention recurrent convolutional network (ARCNet) based on an attention mechanism for scene classification, which focuses on the key areas by squeezing the extracted features into a series of vectors; and Bi *et al.* [27] proposed an attention pooling-based dense connected convolutional network (APDC-Net), which utilizes a spatial attention mechanism and a multilevel supervision strategy to interpret scenes effectively. Much effort has also been devoted to simplifying the redundant parameters while also maintaining a good performance. For example, Cui *et al.* [28] proposed a lightweight CNN classification method based on branch feature fusion (LCNN-BFF) based on the MobileNetV1 architecture, which uses a lightweight branch fusion strategy to improve the computational efficiency; Yu *et al.* [29] proposed BiMobileNet, which removes the last three layers of MobileNetV2 [30] and shares the feature maps to make the model lighter; and Zhang *et al.* [31] introduced SE-MDPMNet based on MobilenetV2, which adds dilated convolution, a multidilation pooling module, and a squeeze-and-excitation (SE) module.

CNN can also be combined with other advanced techniques for scene classification. Zhang *et al.* [32] proposed an ensemble learning framework named the gradient boosting random convolutional network (GBRCN), which combines the feature maps of various CNN backbones (CNet and RCNet) for scene classification; Minetto *et al.* [33] proposed Hydra, which combines multiple ResNet and DenseNet models for training and then conducts majority voting to decide the final label; Li *et al.* [34] proposed a novel adaptive matching discriminant learning network (DLA-MatchNet) for scene classification, which uses the adaptive multilayer perceptron (MLP) to calculate the score of semantic correlation; and Cheng *et al.* [35] proposed a D-CNN model that considers the intraclass diversity and interclass similarity for scene classification. Metric learning is introduced to enhance the recognition ability of the D-CNN model; Ma *et al.* [36] proposed a multiobjective neural evolution automatic search network, which provides a more flexible network architecture for scene classification. To solve the multilabel scene categorization problem, Kang *et al.* [37] proposed a graph relation network (GRN), which models the relations between scenes by making use of a graph structure; Xu *et al.* [38] proposed a deep feature aggregation framework based on the graph convolutional network (GCN) called DFAGCN, which can effectively represent the objects' relevance using graphs.

For remote sensing image scene classification, it is very important to explore both the contextual information and the spatial distribution of the objects contained in the scene, which can express the semantic label of the scene. For example, the traditional pixel- and object-based classification methods can distinguish the basic land-cover types, such as building, vegetation, and water. When the spatial correlations of these objects are considered, scene classification can be used to identify the high-level semantic information, such as industrial area, commercial area, and residential area. CNNs hierarchically extract the local and global features of the imagery by stacking multiple convolutional layers and can achieve good performances in scene classification. However, the global interactions of the local elements in the imagery are not directly modeled when using stacked receptive fields. Differing from CNNs, transformer models directly consider the correlation of the local elements by the use of a self-attention mechanism, which provides a new approach for scene classification. The transformer model [39] was first proposed in 2017 to perform the task of natural language processing (NLP). Subsequently, many variations of the transformer model [40]–[42] have been proposed, which have outperformed some of the state-of-the-art models, such as the recurrent neural networks (RNNs) and the long short-term memory (LSTM) model. The transformer model is also introduced into the hyperspectral image classification task and acquired promising results [43]. In 2020, the transformer model was extended to the vision transformer (ViT) model [44] for image classification. The ViT model uses a ViT encoder to model the correlation of local patches. In 2021, Bazi *et al.* [45] introduced the ViT model [44] into remote sensing image scene classification. The ViT model provides a different way to represent the semantic information of HSR images, by directly considering the contextual information and the spatial distribution of the objects contained in the scene.

However, the traditional ViT model needs to be improved for use with HSR remote sensing images, which contains more complex spectral and spatial information, from three aspects.

1) The traditional ViT model builds the interactions of local patches of a fixed size, which has limitations in preserving the local structural information of ground objects.
2) In the ViT model, a learnable classification token is used as the input feature before the final classifier. However, the channel information in the classification token is not well considered.
3) The multihead self-attention (MSA) mechanism increases the number of parameters. To address the abovementioned problems, a spatial-channel feature preserving ViT (SCViT) model is proposed in this article to better explore the abundant spatial and spectral information contained in HSR remote sensing images.

In this article, the major contributions are given as follows.

1) The performance of the ViT model in remote sensing image scene classification is explored, and the SCViT model is proposed, which improves the discriminative ability of the ViT model by considering both the structural and channel information of HSR imagery. The model parameters are also reduced due to the improved discriminative ability.
2) A progressive aggregation (PA) strategy that iteratively combines the overlapping neighboring tokens is proposed to preserve the local structural information of the HSR imagery.
3) A lightweight channel attention (LCA) module is designed to consider the importance of the different

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feature channels in the classification token before the final classifier.

The rest of this article is organized as follows. The transformer and ViT models are introduced in Section II. In Section III, the proposed SCViT model is described in detail. In Section IV, the experimental results of the proposed model are compared with those of the state-of-the-art scene classification methods on three popular datasets, i.e., UCM, AID, and NWPU datasets. Finally, our conclusions are given in Section V.

II. Related Work

A. Transformer Model

Since 2017, numerous studies based on the transformer model have emerged in the NLP field [39]. For example, Devlin *et al.* [40] proposed a transformer model named bidirectional encoder representations from transformers (BERT), which is pretrained on the unlabeled text and then fine-tuned on smaller datasets for a series of downstream tasks. The generative pretrained transformer (GPT) model [41] and GPT-2 model [42], which rely on a transformer decoder to perform a unique unidirectional pretraining strategy, were also proposed for NLP generation tasks. According to the transformer architecture, transformer models can be divided into three main categories: 1) encoder–decoder architectures; 2) encoder-only architectures; and 3) decoder-only architectures. The encoder–decoder architectures are suitable for sequence-to-sequence modeling, the encoder-only architectures are used in classification or sequence labeling problems, and the decoder-only architectures are utilized in the task of sequence generation. Transformer models rely on a self-attention mechanism. We introduce the theory of the self-attention mechanism in the following.

$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ is the sequence of input vectors, where n is the number of vectors, and the dimension of each vector $\mathbf{x}_i$ is d. The self-attention mechanism models the global interactions of each vector. First, three trainable weight matrices called queries $\mathbf{W}^Q$, keys $\mathbf{W}^K$, and values $\mathbf{W}^V$ are defined and randomly initialized. $\mathbf{X}$ is projected onto the three matrices to acquire $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$. The query vector is matched with the keys vector to measure the similarity by the dot-product operation $\mathbf{QK}^T$ and is divided by the scaled $\sqrt{d_k}$ to achieve normalization and alleviate the problem of the vanishing gradient. A softmax function is then used to obtain the attention matrix. Finally, the weighted sequence $\mathbf{X}'$ is acquired according to the following:

$$\mathbf{X}' = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_q}}\right)\mathbf{V}. \quad (1)$$

A single-head self-attention operation has the limitation of focusing on only one certain position, whereas multiple salient positions appear in real datasets. Instead of simply applying single-head attention, the MSA mechanism is introduced, where the output is obtained by combining the m outputs of the single-head self-attention operations. The input sequence is first separated into m groups. For each group i, the queries $\mathbf{Q}_i$, keys $\mathbf{K}_i$, and values $\mathbf{V}_i$ matrices are defined and randomly initialized. The MSA process can be formulated, as shown in the following:

$$\text{MultiHead}\left(\mathbf{Q}', \mathbf{K}', \mathbf{V}'\right) = \text{Concat}(\text{h}_1, \ldots, \text{h}_\text{h})\mathbf{W}^\mathbf{O} \quad (2)$$

where h_i represents the single-head self-attention, i.e., $\text{Attention}(\mathbf{Q_i}, \mathbf{K_i}, \mathbf{V_i})$, and the single attention head h_i is provided with different matrices for the query vectors $\{\mathbf{Q_i}\}_{i=1}^m$, key vectors $\{\mathbf{K_i}\}_{i=1}^m$, and value vectors $\{\mathbf{V_i}\}_{i=1}^m$. Performing random initialization can result in different representation subspaces for the input vectors. $\mathbf{W}^\mathbf{O}$ is the linear projection matrix.

B. Vision Transformer Model

Since 2020, the transformer model has been gradually applied to computer vision tasks, such as classification [44], segmentation [46], and super-resolution [47]. Dosovitskiy *et al.* [44] proposed a variant transformer for vision tasks called the ViT model. The ViT model removes the transformer decoder for computer vision classification tasks and is typically an encoder-only architecture. In addition, it has been shown that the ViT model is more effective when pretrained on large datasets [48]. In 2021, Bazi *et al.* [45] applied the ViT model [44] to the task of remote sensing scene classification. Multiple image enhancement strategies (CutMix, Cutout, and Hybrid) were utilized, and the experimental results demonstrated the potential and feasibility of transformer-based methods for remote sensing image scene classification.

The process of the ViT model can be expressed as follows. In order to achieve end-to-end ViT training, each input image $\mathbf{X} \in \mathbb{R}^{h\times w\times c}$ is first split into nonoverlapping patches and reshaped into a sequence of flattened fixed-size feature vectors $\mathbf{T}_f \in \mathbb{R}^{n\times(p^2\times c)}$, where (h, w) is the width and height of the image, c is the number of channels, and p is the width of the square patch. $n = (hw/p^2)$ is the length of the input sequence for the ViT encoder. In the ViT model, the split patches are called "tokens." The trainable position embeddings $\mathbf{E}_{\text{pos}}$ are then appended for each token to consider the position information. After this, a learnable classification token $\mathbf{t}_{\text{cls}}$ is added at the front of the sequence, which is utilized in the classification step. The linear embedding process can be formulated, as shown in the following:

$$\mathbf{T}_0 = \left[\mathbf{t}_{\text{cls}}; \mathbf{T}_f\right] + \mathbf{E}_{\text{pos}}, \quad \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(n+1)\times d}. \quad (3)$$

The sequence is then fed into the ViT encoder to consider the interactions of each token. The encoder contains several ViT blocks. Each block consists of one MSA layer and a two-layer MLP as follows:

$$\begin{aligned}\mathbf{T}'_\ell &= \text{MSA}(\text{LN}(\mathbf{T}_{\ell-1})) + \mathbf{T}_{\ell-1}, \quad \ell = 1, \ldots, L \\ \mathbf{T}_\ell &= \text{MLP}\left(\text{LN}\left(\mathbf{T}'_\ell\right)\right) + \mathbf{T}'_\ell, \quad \ell = 1, \ldots, L \end{aligned} \quad (4)$$

where $\mathbf{T}_{\ell-1}$ is the output token sequence acquired in block $\ell-1$, $\mathbf{T}'_\ell$ is the sequence generated after the MSA mechanism, $\mathbf{T}_\ell$ is the output sequence of block ℓ, and LN represents the layer normalization. Finally, a fully connected layer followed by a softmax classifier is used to classify the classification

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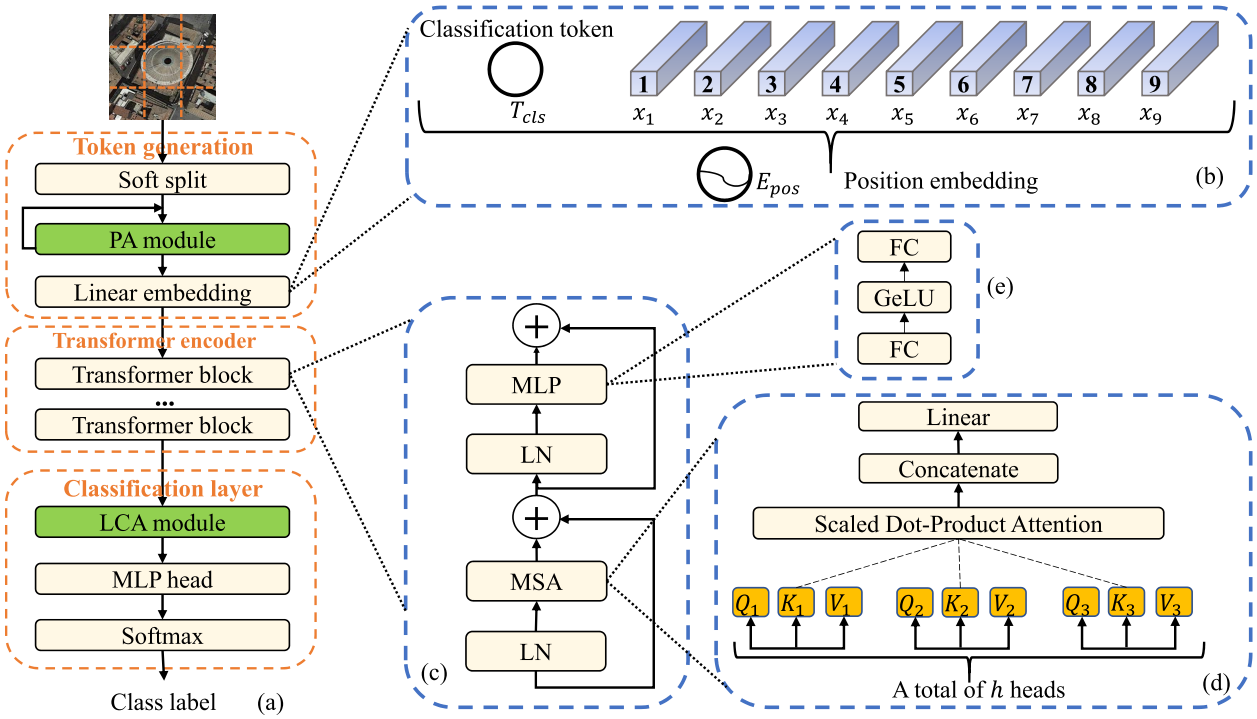


Fig. 1. (a) Proposed SCViT model consists of three main parts: token generation, the ViT encoder, and the classification layer. (b) Details of the linear embedding. (c) Overview of the ViT block. (d) MSA mechanism. (e) Structure of MLP.

token $\mathbf{t}_{\text{cls}}$ learned from the ViT encoder to acquire the scene categories as follows:

$$\mathrm{y} = \mathrm{softmax}(\mathrm{fc}(\mathrm{LN}(\mathbf{t}_{\text{cls}}))). \tag{5}$$

III. Methodology

The flowchart of the SCViT model is displayed in Fig. 1. The proposed SCViT model consists of three main parts: 1) token generation (the PA strategy and linear embedding); 2) the ViT encoder; and 3) a classification layer. In the first part, the original image is split into several overlapping patches by a sliding window to acquire the initial tokens. New tokens are then generated by the PA module, so as to consider the local contextual information. A trainable classification token is then added, which is used to perform the classification task. Position embedding is also performed for each token to preserve the local spatial information, which is called linear embedding. Details of the token generation are provided in Section III-A. In the second part of the SCViT model, the encoder contains several ViT blocks. In each block, an MSA mechanism followed by a two-layer MLP is used to model the interactions of each token. The ViT encoder is introduced in detail in Section III-B. In the third part of the model, the output classification token in the encoder is classified, and the LCA module is added to consider the channel weight in the classification token. The details of the third part of the model are provided in Section III-C.

A. Token Generation

1) Progressive Aggregation Module: In this part, the PA strategy [49] is introduced to make the tokens preserve more local spatial information. As shown in Fig. 2, there are two stages (reshape and split) contained in the PA module. In the first stage, the initial tokens are reshaped into a 3D image cube (width, height, and channel) that contains the local contextual information. In the second stage, the 3D cube is first split into some overlapping patches. New tokens are then acquired by connecting the neighboring patches.

$\mathbf{T}_i'$ represents the series of tokens generated from the last iteration, where i represents the ith iteration. In the reshape step, the sequence $\mathbf{T}_i'$ is reshaped into a 3D image cube $\mathbf{I}_i \in \mathbb{R}^{h\times w\times c}$, where w and h represent the width and height of the cube. Note that $\mathbf{T}_0'$ is the sequence generated from the original image.

In the split step, the image cube $\mathbf{I}_i$ is first split into several overlapping patches by a sliding window, and every four adjacent tokens are connected to acquire a new token, as shown in Fig. 2 (e.g., [1,2,4,5], [4,5,7,8], ..., [5,6,8,9]). The overlapping patches are used to better consider the complete boundary information of objects with different scales. The splitting and connecting operation can be formulated, as shown in the following equation:

$$\mathbf{T}_{i+1} = \mathrm{Split}(\mathbf{I}_i), \quad i = 1, \ldots, (n-1) \tag{6}$$

where n is the number of iterations and $\mathbf{T}_{i+1}$ is the sequence of tokens after the split. $k \times k$, p, and $k - s$ are the window size, the padding, and the stride, respectively. The length of the sequence $\mathbf{T}_{i+1}$ can be calculated, as shown in the following equation:

$$\begin{aligned} l_{i+1} &= H_{i+1} \times W_{i+1}, \quad i = 1, \ldots, (n-1) \\ H_{i+1} &= \left\lfloor \frac{h_i + 2p_i - k_i}{k_i - s_i} + 1 \right\rfloor \\ W_{i+1} &= \left\lfloor \frac{w_i + 2p_i - k_i}{k_i - s_i} + 1 \right\rfloor. \end{aligned} \tag{7}$$

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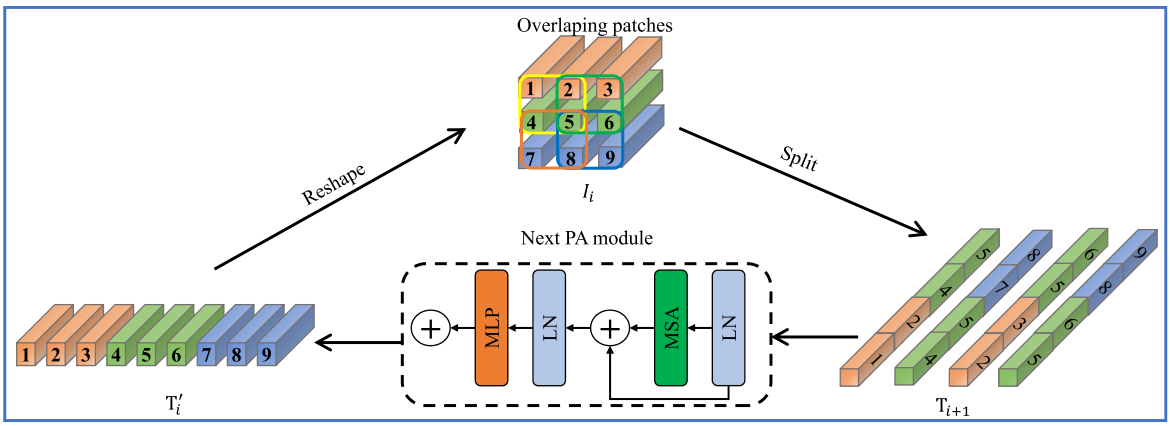


Fig. 2. PA module, including the reshape and split stages.

The MSA operation and MLP with skip connection are then performed on the sequence $\mathbf{T}_{i+1}$ so that the values of the tokens can be adjusted in the token generation part, which is better for the classification, as follows:

$$\mathbf{T}'_{i+1} = \mathrm{MLP}(\mathrm{MSA}(\mathbf{T}_{i+1})). \tag{8}$$

2) Linear Embedding: $\mathbf{T}_f$ represents the output sequence of tokens acquired from the PA module. A trainable classification token $\mathbf{t}_{\mathrm{cls}}$ is first added at the front of the sequence $\mathbf{T}_f$ and is used to perform the final classification task in the ViT encoder [40]. Thus, the length of the sequence becomes $l+1$. A position embedding channel $\mathbf{E}_{\mathrm{pos}}$ is then added for each token to keep the relative spatial position information. The linear embedding process can be formulated, as shown in the following:

$$\mathbf{T}_{f_0} = \left[\mathbf{t}_{\mathrm{cls}}; \mathbf{T}_f\right] + \mathbf{E}_{\mathrm{pos}} \tag{9}$$

where $\mathbf{T}_{f_0}$ is the final output of the token generation step before the encoder.

B. ViT Encoder

In this part, as displayed in Fig. 1(c), the token sequence $\mathbf{T}_{f_0}$ is input into the ViT encoder, which contains several ViT blocks, to model the interactions of the tokens in $\mathbf{T}_{f_0}$ and acquire the classification token $\mathbf{t}_{\mathrm{cls}}$ for the third part.

Each block consists of an MSA layer and a two-layer MLP. Layer normalization is also added to improve the robustness of the model. The ViT blocks are then formulated, as shown in the following equation:

$$\mathbf{T}_{f_i} = \mathrm{MLP}\left(\mathrm{MSA}\left(\mathbf{T}_{f_i-1}\right)\right), \quad i = 1, \ldots, b \tag{10}$$

where b indicates the number of ViT blocks.

C. Classification Layer

In this part, the classification token $\mathbf{t}_{\mathrm{cls}}$ in the token sequence $\mathbf{T}_{f_b}$ is selected to perform the final scene classification task. An LCA module is also introduced to consider the weight of the channels in the classification token $\mathbf{t}_{\mathrm{cls}}$. The weighted token is then classified by the MLP head, followed by the softmax classifier, to acquire the scene label y, as follows:

$$\mathrm{y} = \mathrm{softmax}(\mathrm{fc}(\mathrm{LCA}(\mathbf{t}_{\mathrm{cls}}))). \tag{11}$$

In the ViT encoder, the global interaction of the local tokens is modeled by the MSA mechanism. However, the MSA mechanism mainly considers the information in the spatial domain, but the channel information is also important for classification. The LCA module considers the channel weights in the classification token to improve the classification accuracy. The LCA module replaces the fully connected layer used in some popular channel attention modules, such as the SE module [50], with 1D convolution, to achieve fewer parameters while maintaining a good performance.

As shown in Fig. 3, the classification token $\mathbf{t}_{\mathrm{cls}} \in \mathbb{R}^{p\times c}$ represents the input state. First, global average pooling (GAP) is utilized to reduce the spatial dimension of $\mathbf{t}_{\mathrm{cls}}$ and acquire a feature vector $\mathbf{f} \in \mathbb{R}^{1\times c}$. Second, the channel weight ω is calculated using the following operation:

$$\omega = \sigma\left(\mathrm{Conv1\ D}_k(\mathbf{t}_{\mathrm{cls}})\right) \tag{12}$$

where Conv1d is the 1D convolutional operation with the kernel size k, where the number of kernels is the same as the channels of the classification token, and σ is a sigmoid function. Finally, the reweighted token $\bar{\mathbf{t}}_{\mathrm{cls}}$ is acquired as follows:

$$\bar{\mathbf{t}}_{\mathrm{cls}} = \mathbf{t}_{\mathrm{cls}} \cdot \omega. \tag{13}$$

IV. Experiment and Analysis

A. Experiment Settings

In order to verify the performance of the proposed SCViT model, we chose three popular datasets: 1) the UC Merced (UCM) dataset [51]; 2) the Aerial Image Dataset (AID) [52]; and 3) the NWPU-RESISC45 (NWPU) dataset [5]. Some scene image samples from the above datasets are shown in Figs. 4–6. The three datasets are all commonly used benchmarks, which have the following challenges.

1) The three datasets represent small, medium, and large data quantities, respectively.
2) The images in the datasets are acquired from different sensors with complex spectral and spatial distributions. Specifically, the AID and NWPU datasets are multi-source remote sensing datasets, which are more difficult

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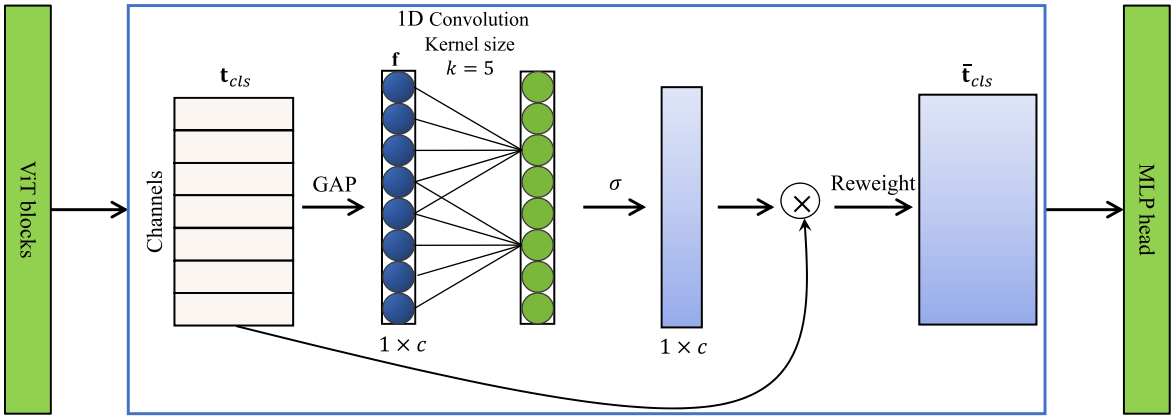


Fig. 3. Structure of the LCA module.



Fig. 4. Sample images for each class in the UCM dataset.

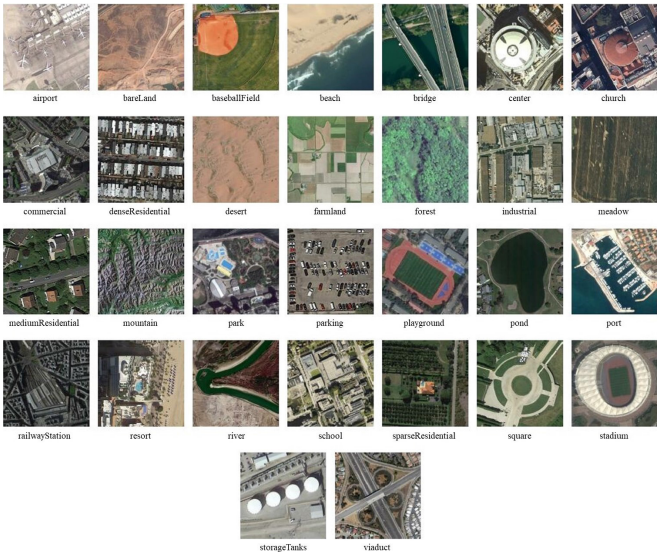


Fig. 5. Sample images for each class in the AID dataset.

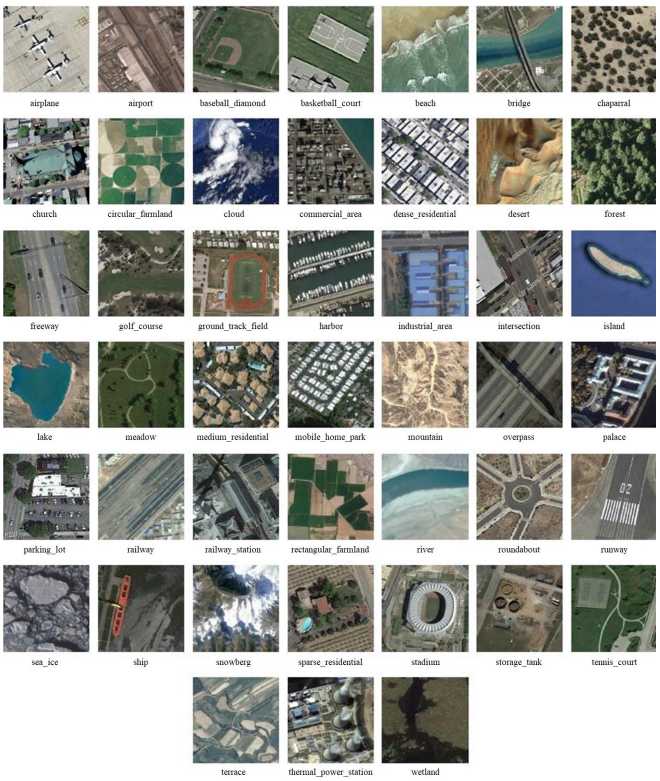


Fig. 6. Sample images for each class in the NWPU dataset.

to classify than single-source datasets, such as the UCM dataset.

3) The categories of the three datasets vary a lot.

Each experimental dataset was divided into two parts, where the training ratios for the UCM dataset (80%, 50%) and AID (20%, 50%) were selected according to [52]. The training ratios for the NWPU dataset (10%, 20%) were set according to [5]. In order to prevent overfitting in the training stage and improve the generalization ability of the models, data augmentation was utilized. The image augmentation methods included fixed image rotation, random flipping vertically and horizontally, random rotation by 0°–20°, random clipping and padding with 16 pixels, illumination adjustment, and so on.

In the following, we group the comparison methods according to the algorithm type. For the CNN-based methods, we selected the following classical CNN-based methods: CaffeNet [52], AlexNet [5], GoogLeNet [52], VGGNet-16 [52], and ResNet [53]. For the attention-based CNN methods, we selected ADFF [25], ARCNet [26], APDC-Net [27], and ResNet-50+EAM [53]. For the lightweight CNN-based methods, we selected MobileNetV2 [31], SE-MDPMNet [31], LCNN-BFF [28], and BiMobileNet [29]. The performances of

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TABLE I
OA (%) FOR THE DEEP LEARNING-BASED METHODS ON THE UCM DATASET

Model	Year	OA(%)	
		50%	80%
CaffeNet [52]	2017	93.98±0.67	95.02±0.81
GoogLeNet [52]	2017	92.70±0.60	94.31±0.89
VGG-VD-16 [52]	2017	94.14±0.69	95.21±1.20
Fine-tune ResNet-50 [53]	2020	-	98.69±0.49
ADFF [25]	2019	96.05±0.56	97.53±0.63
ARCNet [26]	2019	96.81±0.14	99.12±0.40
APDC-Net [27]	2020	95.01±0.43	97.05±0.43
Fine-tune MobileNetV2 [31]	2018	97.88±0.31	98.13±0.33
SE-MDPMNet [31]	2018	98.57±0.11	98.95±0.12
LCNN-BFF [28]	2020	94.64±0.21	99.29±0.24
BiMobileNet [29]	2020	98.45±0.27	99.03±0.28
D-CNN with VGGNet-16 [35]	2018	-	98.93±0.10
DFAGCN [38]	2021	-	98.48±0.42
V16_21k [45]	2021	98.14±0.47	-
ViT [44]	2021	98.48±0.39	99.01±0.21
ViT+PA	2021	98.58±0.31	99.14±0.27
ViT+LCA	2021	98.68±0.26	99.43±0.31
ViT+SE	2021	98.55±0.20	99.10±0.20
ViT+PA+SE	2021	98.81±0.30	99.42±0.23
SCViT	2021	**98.90±0.19**	**99.57±0.31**

TABLE II
OA (%) FOR THE DEEP LEARNING-BASED METHODS ON THE AID DATASET

Model	Year	OA(%)	
		20%	50%
CaffeNet [52]	2017	86.86±0.47	89.53±0.31
GoogLeNet [52]	2017	83.44±0.40	86.39±0.55
VGG-VD-16 [52]	2017	86.59±0.29	89.64±0.36
Fine-tune ResNet-50 [53]	2020	93.64±0.25	96.62±0.13
ADFF [25]	2019	93.68±0.29	94.75±0.25
ARCNet [26]	2019	88.75±0.40	93.10±0.55
APDC-Net [27]	2020	88.56±0.29	92.15±0.29
Fine-tune MobileNetV2 [31]	2018	94.13±0.28	95.96±0.27
SE-MDPMNet [31]	2018	94.68±0.17	**97.14±0.15**
LCNN-BFF [28]	2020	91.66±0.48	94.62±0.16
BiMobileNet [29]	2020	94.83±0.24	96.87±0.23
D-CNN with VGGNet-16 [35]	2018	90.82±0.16	96.89±0.10
DFAGCN [38]	2021	-	94.88±0.22
V16_21k [45]	2021	94.97±0.01	-
ViT [44]	2021	94.90±0.29	96.49±0.18
ViT+PA	2021	95.31±0.11	96.72±0.16
ViT+LCA	2021	95.43±0.22	96.89±0.41
ViT+SE	2021	95.43±0.22	95.61±0.57
ViT+PA+SE	2021	95.49±0.15	96.81±0.47
SCViT	2021	**95.56±0.17**	96.98±0.16

some of the advanced networks, i.e., the discriminative CNN (D-CNN) method [35] and the GCN [38], were also compared. The hyperparameters were selected according to the original papers.

For the ViT-based methods, the original ViT model [44], V16_21K model [45], and the proposed SCViT are compared. We also introduce the following ablation experiments.

1) *ViT+PA:* The ViT model with the PA module.
2) *ViT+LCA:* The ViT model with the LCA module.
3) *ViT+SE:* The ViT model with an SE module.
4) *ViT+PA+SE:* The ViT model with the PA module and an SE module.

The ViT-based methods were first pretrained on the ImageNet dataset. The methods were then fine-tuned on the remote sensing datasets with 60 learning epochs and a batch size of 64. Cosine learning and the AdamP [54] optimizer were used as the training strategy. The initial learning rate was 0.01, and the minimum learning rate was 0.0001. The experiments were repeated five times, and the datasets were randomly shuffled at the beginning of the experiments. The overall accuracy (OA) and the confusion matrix are used in this article as the main quantitative measures.

The methods in the experimental part were implemented in an Ubuntu 18.04.5 LTS operating system with an Intel[1] Xeon[2] E5-2650 v4 central processing unit (CPU) @ 2.20 GHz and 128 GB of memory. An NVIDIA Tesla V100 SXM2 high-speed graphics processing unit (GPU) with 32 GB of memory was also used.

B. Experiment 1: UCM Dataset

The UCM dataset was published by the University of California at Merced in 2010, where the images were downloaded from the USGS National Map Urban Area Imagery collection. As one of the most widely used datasets in the task of remote sensing image scene classification, the UCM dataset covers most regions of the United States. This dataset contains 21 scene classes (see Fig 4) with 100 images per class. The spatial size of each image is 256×256 pixels. The images contain three bands, i.e., red, green, and blue.

The quantitative results for the UCM dataset are displayed in Table I. The best results have been highlighted in bold. Among the classical CNN methods, ResNet acquires the best result, and it can be regarded as the representative baseline for the classical CNN models. When an attention mechanism is considered, the attention-based CNN models achieve higher accuracies than the classical CNN models. The OA of the original ViT model is 99.01% at an 80% training ratio, which is comparative to the result of ResNet. Overall, the results confirm the potential of the ViT models. When the local structure of the image is considered, the ViT+PA model acquires better results at the ratios of both 50% and 80% than the original ViT model. The results of ViT+LCA indicate that the channel attention mechanism is important for the ViT-based methods. The results of ViT+SE and ViT+LCA show the superiority of the proposed LCA module over the SE module. The accuracy of the proposed SCViT model at the ratio of 50% is 98.90%, which is the best result obtained on the UCM dataset. At the ratio of 80%, the SCViT model also acquires better results than ViT+PA and ViT+LCA.

The confusion matrices for the ViT and SCViT models on the UCM dataset with an 80% training ratio are displayed in Fig. 7(a) and (b) to verify the performance of the PA module and the LCA module on the categories with complex spatial distributions. The diagonal elements in Fig. 7(a) show that the ViT model acquires a good performance in most categories of

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the UCM dataset, except dense residential. From the samples of dense residential (see Fig. 4), it can be seen that this category contains images with densely distributed buildings, indicating complex spatial structures. As shown in Fig. 7(b), the SCViT model acquires better results in each category and, especially, the class of dense residential compared with the original ViT model. These results indicate that the proposed SCViT model has a more powerful discriminative ability in scenes with complex spatial information.

C. Experiment 2: AID Dataset

The AID dataset was published by Wuhan University in 2016. The AID dataset is a multisource dataset, which is made up of images from Google Earth of different remote sensing sensors. The AID dataset covers different regions of many countries, including China, Japan, France, and the United States. The dataset contains 30 scene classes (see Fig. 5) with 200–400 images per class, totaling 10 000 images. The size of each image is 600×600, and the spatial resolution of the images varies from 0.5 to 0.8 m. The dataset also contains RGB channels.

The results for the AID dataset are listed in Table II. Compared with the classical CNNs, the original ViT model acquires good results under the training ratios of both 20% and 50%. The accuracies of the original ViT model reach 94.90% and 96.49%, which are promising results for the AID dataset. Among the ViT-based models, the V16_21k model with data enhancement shows some improvement over the original ViT model. When only considering local structural information, the OA of the ViT+PA model reaches 95.31% and 96.72%. When only considering the importance of different channels, the OA of the ViT+LCA model reaches 95.43% and 96.89%. The proposed SCViT model acquires the best performance on the AID dataset because not only the local structure but also the channel information is well explored by the proposed model. Note that the AID dataset contains more categories than the UCM dataset so that the proposed SCViT model acquires greater improvements at a lower percentage of training samples (20%). The variance of the SCViT model is also lower than that of the original ViT model, which indicates that the proposed model is more stable than the original ViT model on this dataset.

The confusion matrices for the ViT and SCViT models on the AID dataset are displayed in Fig. 7(c) and (d), for which the training ratio was set to 50%. From Fig. 7(c), it is apparent that the classes of center, church, and commercial are easily confused. As can be seen in Fig. 5, the spatial and spectral characteristics of these three classes are similar. Although the original ViT model has difficulty in classifying these classes, as shown in Fig. 7(d), the proposed SCViT model acquires better results on all three classes, which indicates that the SCViT model has the ability to consider the complex spatial features contained in HSR images. The resort class is another class, which is easily misclassified. Compared with the original ViT model, the SCViT model shows an obvious improvement in the resort class. From the diagonal elements, it can be seen that the proposed SCViT model obtains good results on most categories. However, there are still some classes that are difficult for the SCViT model to classify, such as square and school. It can be seen in Fig. 5 that the square class has few distinct features, which makes it difficult to classify.

TABLE III
OA (%) FOR THE DEEP LEARNING-BASED METHODS ON THE NWPU DATASET

Model	Year	OA(%)	
		10%	20%
Fine-tune AlexNet [5]	2017	81.22±0.19	85.16±0.18
Fine-tune VGGNet-16 [5]	2017	87.15±0.45	90.36±0.18
Fine-tune GoogLeNet [5]	2017	82.57±0.12	86.02±0.18
Fine-tune ResNet-50 [53]	2020	88.48±0.21	91.86±0.19
ADFF [25]	2019	90.58±0.19	91.91±0.23
APDC-Net [27]	2020	85.94±0.22	87.84±0.26
ResNet-50+EAM [53]	2020	90.87±0.15	93.51±0.12
Fine-tune MobileNetV2 [31]	2018	90.16±0.12	93.00±0.18
SE-MDPMNet [31]	2018	91.80±0.07	94.11±0.03
LCNN-BFF [28]	2020	86.53±0.15	91.73±0.17
BiMobileNet [29]	2020	92.06±0.14	94.08±0.11
D-CNN with VGGNet-16 [35]	2018	89.22±0.50	91.89±0.22
DFAGCN [38]	2021	-	89.29±0.28
V16_21k [45]	2021	92.60±0.10	-
ViT [44]	2021	91.59±0.19	93.90±0.20
ViT+PA	2021	92.65±0.20	94.24±0.16
ViT+LCA	2021	92.37±0.20	94.63±0.13
ViT+SE	2021	92.39±0.21	94.60±0.17
ViT+PA+SE	2021	92.42±0.26	94.64±0.15
SCViT	2021	**92.72±0.04**	**94.66±0.10**

D. Experiment 3: NWPU Dataset

The NWPU dataset was published by Northwestern Polytechnical University, China, in 2016. The NWPU dataset is also a multisource dataset, which contains a large number of remote sensing images and scene categories. The dataset contains 45 carefully selected and meaningful scene classes (see Fig. 6), with 700 images per class, totaling 31 500 images. The spatial resolution of each image varies from 0.2 to 30 m, and the size of each image is 256×256. The NWPU dataset is a very challenging and extensively used dataset. Although the NWPU dataset does not have the problem of imbalanced categories, which is a problem in the AID dataset, the NWPU dataset contains many subclasses, which are easily misclassified. For example, the residential class is further divided into dense_residential, medium_residential, and sparse_residential, which increases the difficulty of the classification. Another challenge with the NWPU dataset is that the spatial resolution of the images varies a lot.

In this experiment, the training ratios were set to 10% and 20%. The results of the experimental methods on the NWPU dataset are listed in Table III. From Table III, it can be seen that the attention-based CNN methods obtain better results than the traditional CNN methods, which indicates that considering the importance of local regions and channels is effective on this dataset. The lightweight CNN models also acquire satisfactory results. For the ViT-based methods, the original ViT model acquires OAs of 91.59% and 93.90% at 10% and 20%, respectively, which are better results than those obtained by the classical CNNs. The V16_21k model

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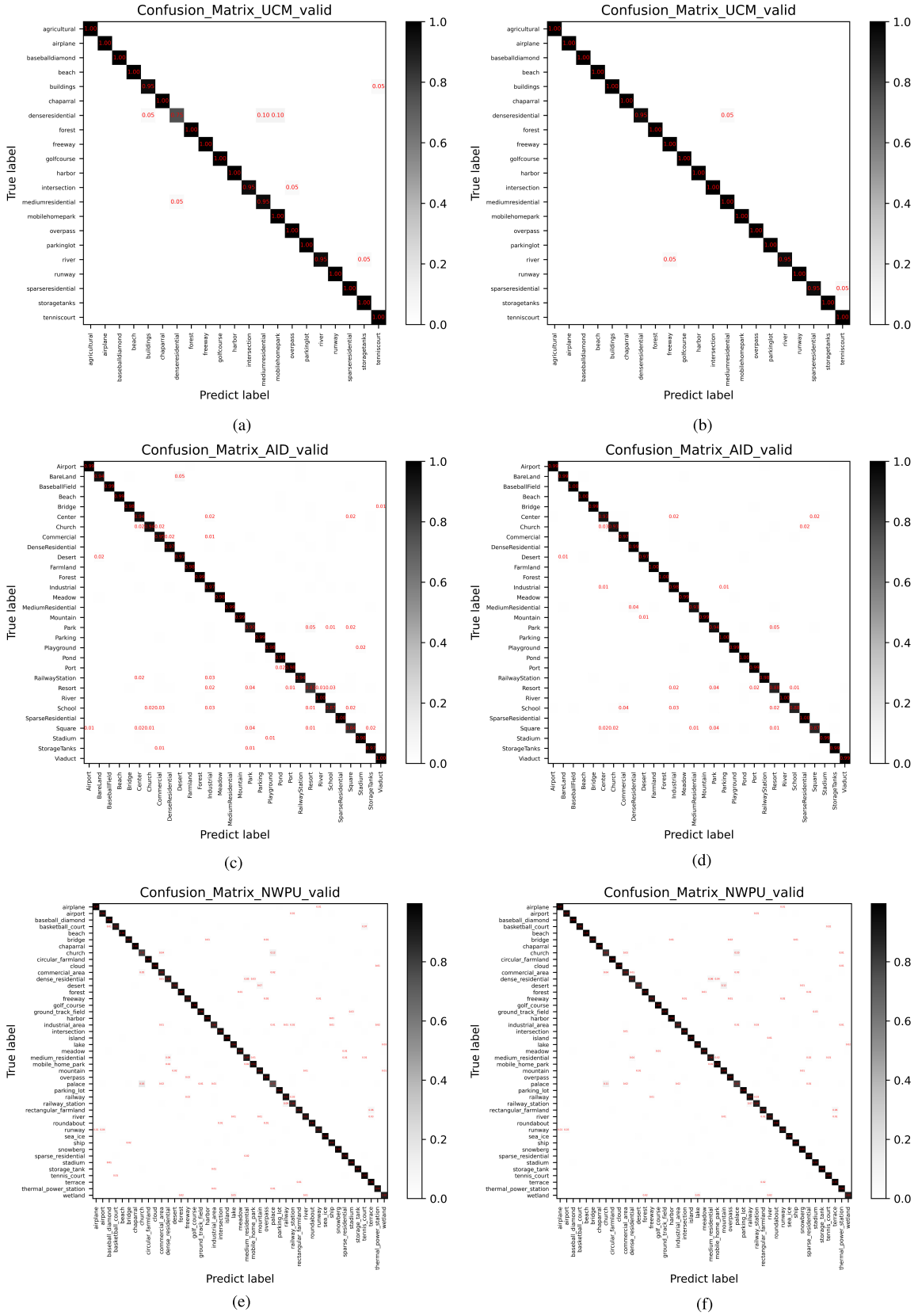


Fig. 7. Confusion matrices for the UCM dataset (with a training ratio of 80%), the AID dataset (with a training ratio of 50%), and NWPU dataset (with a training ratio of 20%) obtained by the ViT and SCViT models. (a) ViT-UCM. (b) SCViT-UCM. (c) ViT-AID. (d) SCViT-AID. (e) ViT-NWPU. (f) SCViT-NWPU.

with data augmentation achieves higher accuracy than the original ViT model at a ratio of 10%. When considering the local structure, the ViT+PA model also obtains better results at 10% and 20%. The LCA module also results in

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TABLE IV
HYPERPARAMETERS FOR THE LCA MODULE KERNEL SIZE

Dataset (training rate)	OA(3*3)	OA(5*5)	OA(7*7)
UCM(50%)	99.04	98.86	98.57
UCM(80%)	99.50	99.76	99.04
AID(20%)	95.06	95.42	95.00
AID(50%)	97.18	97.30	97.18
NWPU(10%)	91.81	92.57	89.06
NWPU(20%)	94.30	94.77	94.30

TABLE V
MODELS' PARAMETERS

Model	Number of ViT blocks	MLP size	Heads
ViT-S [44]	8	2358	12
ViT-B [44]	12	3072	16
ViT-L [44]	24	4096	16
SCViT-S	8	512	4
SCViT-B	14	1152	6
SCViT-L	19	1344	7

TABLE VI
PARAMETERS AND GMACS OF THE DEEP LEARNING-BASED METHODS ON THE NWPU DATASET

Model	Parameters	GMACs	GPU time	OA (20%)
VGGNet-16 [5]	138.36M	7.75G	4.744ms	90.36
GoogLeNet [5]	7M	0.75G	24.351ms	86.02
ResNet-50 [53]	25.56M	4.14G	30.895ms	91.86
MobileNetV2 [31]	3.5M	0.35G	20.183ms	90.16
ViT-S [44]	49M	10.1G	18.921ms	93.92
ViT-B [44]	86M	17.6G	22.084ms	94.13
ViT-L [44]	304M	63.6G	27.074ms	94.38
SCViT-S	4.5M	1.3G	21.848ms	94.75
SCViT-B	22.1M	6.3G	27.141ms	94.85
SCViT-L	40.1M	9.3G	33.681ms	**94.99**

big improvements on this dataset. The proposed SCViT model acquires better results than the other ViT models. The variance of the SCViT model is also the lowest, which indicates that it is more robust than the other models on the NWPU dataset.

The confusion matrices for the ViT and SCViT models are displayed in Fig. 7(e) and (f) for which the training ratio was set to 20%. From the diagonal elements, the ViT model performs well in most classes, and the average OA reaches 90%. However, there are still some difficult classes, such as church and palace. We can see from Fig. 6 that the architecture and style of the two classes are similar. The original ViT model obtains OAs of only 79% and 73% on these classes, whereas the SCViT model shows a big improvement with the OA reaching 83% and 80%. The SCViT model also performs better in the industrial_area and railway_station classes. However, there are still several classes that the SCViT model struggles with, such as commercial_area, dense_residential, and wetland.

From the results of the three experiments, we can make the following conclusions.

1) Compared with the CNN-based models, the ViT models obtain superior results, confirming the potential of the ViT models for scene classification.
2) When the training ratio is increased, the ViT models obtain better performances, which shows that the ViT models need adequate training data to acquire good results.
3) Compared with the original ViT model, the proposed SCViT model shows superior performance on scene classes with complex spatial features.

E. Evaluation of the Models

In this part, the hyperparameters and the computational efficiency of the proposed method are analyzed. First, we analyze the impact of the convolutional kernel size in the LCA module. The OAs of the SCViT model under different kernel sizes on the three datasets are listed in Table IV. Table IV indicates that the 5*5 kernel gives the best performances on these datasets. Second, the running efficiency of the proposed SCViT model is analyzed. The model parameters, giga multiply–accumulate operations per second (GMACs), and the time cost of the GPU are used to evaluate the models. We used the NWPU dataset to perform the evaluation as it is the most complex among the three datasets. The structures of the different ViT models are listed in Table V, and the results are listed in Table VI. From Table VI, it can be seen that the classification accuracy of the ViT models increases as the model becomes larger. The running efficiency of the model decreases as the parameters increase. The SCViT model with a large size acquires the best OA but the lowest efficiency. However, the SCViT model with a small size can obtain good results compared with some of the classical CNN models and even some of the lightweight CNN models. These results indicate that the ViT-based models have the potential to be used for the task of scene classification, and the model size could be further reduced.

V. CONCLUSION

In this article, we have investigated the ViT models and compared them with CNNs for high-resolution remote sensing image scene classification. We have also proposed the SCViT model to improve the discriminative ability of the traditional ViT model. First, a PA strategy is introduced, which combines the neighborhood tokens to extract more local structural information of the imagery. Second, the classification ability of the ViT model is improved by introducing an LCA module, which considers the contribution of the different channels contained in the classification token. The experimental results obtained on popular scene classification datasets confirmed the potential of the ViT models for scene classification and, especially, the potential contribution of the proposed SCViT model, which showed superior performance in the experiments conducted in this study. The study results indicate the potential of using the self-attention mechanism to directly build the high-level semantic information contained in HSR remote sensing imagery. The ViT model can play the role of the backbone for other applications, such as segmentation and detection, in the future.

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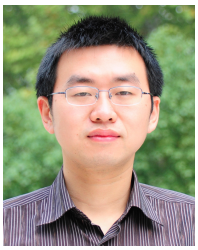
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